Feature Engineering, Model Optimization & Performance Comparison

Artificial Intelligence & Machine Learning – Task 2 Report

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1. Introduction

Machine Learning models often perform differently on the same dataset. Therefore, comparing multiple models and selecting the best-performing one is an important step in the machine learning workflow. In this project, feature engineering and model optimization techniques were applied to the California Housing Dataset to improve prediction performance.

The project focuses on comparing three regression algorithms: Linear Regression, Ridge Regression, and Decision Tree Regressor. Feature scaling was performed using StandardScaler, and the models were evaluated using RMSE and R^2 Score. Various visualizations were also generated to understand feature relationships and model behavior.

The objective of this project is to identify the most effective regression model for house price prediction while gaining practical experience in feature engineering, model evaluation, and performance comparison.

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2. Objectives

The main objectives of this project are:

- To understand the importance of feature engineering.
- To apply feature scaling using StandardScaler.
- To train and compare multiple regression models.
- To evaluate model performance using standard regression metrics.
- To visualize prediction results and model errors.
- To identify the best-performing model for house price prediction.

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3. Dataset Description

The California Housing Dataset from the scikit-learn library was used for this project. The dataset contains information collected from California housing districts and is commonly used for regression tasks.

Features Used

Feature	Description
MedInc	Median income in the area
HouseAge	Average age of houses
AveRooms	Average number of rooms
AveBedrms	Average number of bedrooms
Population	Population of the district
AveOccup	Average occupancy
Latitude	Latitude coordinate
Longitude	Longitude coordinate

Target Variable

HousePrice (Median House Value)

The target variable represents the median house value for each district and is used for prediction.

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4. Exploratory Data Analysis (EDA)

Before training the models, exploratory data analysis was performed to understand the structure and characteristics of the dataset.

The following steps were carried out:

- Dataset shape analysis
- Feature inspection
- Statistical summary generation
- Missing value analysis
- Correlation analysis
- Data visualization

Key Observations

- The dataset contains only numerical features.
- No significant missing values were present.
- Median Income showed the strongest positive correlation with house prices.

- Population exhibited a weaker relationship with house prices.
- Geographic features such as Latitude and Longitude influenced house values.

A correlation heatmap was generated to visualize relationships among all variables.

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5. Feature Engineering and Data Preprocessing

Feature engineering is a critical step in machine learning because models often perform better when features are appropriately transformed.

The following preprocessing steps were performed:

Feature Selection

The target variable was separated from the independent variables.

Feature Scaling

StandardScaler was applied to normalize feature values.

Benefits of feature scaling include:

- Improved model convergence
- Better numerical stability
- Fair comparison among features
- Enhanced model performance

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

This ensures that model performance is evaluated on unseen data.

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6. Model Training

Three machine learning algorithms were implemented and compared.

6.1 Linear Regression

Linear Regression serves as a baseline model for regression tasks. It assumes a linear relationship between input features and the target variable.

Advantages:

- Simple and interpretable
- Fast training
- Easy implementation

6.2 Ridge Regression

Ridge Regression is an extension of Linear Regression that includes regularization.

Advantages:

- Reduces overfitting
- Handles multicollinearity
- Improves generalization

6.3 Decision Tree Regressor

Decision Tree Regressor creates a tree-based structure to make predictions.

Advantages:

- Captures nonlinear relationships
- Handles complex patterns
- Easy to visualize and interpret

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7. Model Evaluation

To compare model performance, two regression metrics were used.

Root Mean Squared Error (RMSE)

RMSE measures the average magnitude of prediction errors.

Characteristics:

- Lower RMSE indicates better performance.
- Penalizes larger prediction errors.

R^2 Score

R^2 Score measures how well the model explains variation in the target variable.

Characteristics:

- Values closer to 1 indicate better performance.
- Higher R^2 Score signifies a stronger predictive model.

The performance of all three models was compared using these metrics.

A comparison table and visualization charts were generated to identify the best-performing model.

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8. Visualization and Analysis

Several visualizations were created to understand data patterns and model performance.

Correlation Heatmap

Used to identify relationships between variables.

Population vs House Price Scatter Plot

Helped analyze the impact of population on house prices.

Median Income vs House Price Scatter Plot

Showed a strong positive relationship between income and housing value.

Actual vs Predicted Plot

Compared predicted values with actual house prices.

Residual Distribution Plot

Analyzed prediction errors and their distribution.

Residual Scatter Plot

Verified whether prediction errors were randomly distributed around zero.

These visualizations provided valuable insights into both the dataset and model behavior.

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9. Performance Comparison

The trained models were compared using RMSE and R^2 Score.

The comparison process helped identify:

- Which model produced the lowest error.
- Which model explained the highest variance.
- Which model generalized better to unseen data.

The model with the highest R^2 Score and lowest RMSE was selected as the best-performing model.

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10. Improvement Ideas

Several improvements can further enhance prediction accuracy:

- Hyperparameter tuning using GridSearchCV
- Cross-validation for robust evaluation
- Outlier detection and removal
- Feature engineering with additional variables
- Ensemble learning methods
- Random Forest Regression
- Gradient Boosting
- XGBoost Regression

These advanced techniques may improve performance beyond traditional regression models.

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11. Conclusion

In this project, feature engineering and model optimization techniques were applied to the California Housing Dataset. Feature scaling was performed using StandardScaler, and three regression models—Linear Regression, Ridge Regression, and Decision Tree Regressor—were trained and evaluated.

The models were compared using RMSE and R^2 Score, and visual analysis was conducted through various plots and graphs. The comparison process provided valuable insights into the strengths and limitations of different regression algorithms.

This project enhanced understanding of feature engineering, model evaluation, performance comparison, and machine learning best practices. It also demonstrated the importance of selecting the most appropriate model for a prediction problem based on objective evaluation metrics.